

An adaptive smart system to foster disabled and elderly people in kitchen-related task

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ABSTRACT

The present paper proposes a new adaptive smart kitchen environment able to support different users with several typologies of impairment (i.e., visual, motor, cognitive) in performing cooking activities. This system is managed through an adaptive user interface, which guides the user in food preparation according to users' capabilities and needs and permits household appliances controlling. Results of a preliminary usability evaluation has shown that the proposed solution is accessible and highly usable for users with some limitations such elderly with mild to moderate dementia and adult people with moderate retinopathy and rheumatoid arthritis.

CCS Concepts

- Human-centered computing → User interface design
- Human-centered computing → Usability testing.

Keywords

Adaptive User Interface, Design for all, Human Computer Interaction, Smart Environment

1. INTRODUCTION

Smart Environment is a significant challenge for current ICT research: it represents one of the solutions which allow elderly and users with impairment to remain in their home as long as possible in a normal daily life environment. In this context, the development of smart kitchen reveals a great interest, as motivated by a critical number of studies [1,2]. One of the strong motivation of such interest is due to the fact that, supporting users in the main activities they perform in the kitchen (i.e., preparing meals, using household appliances) means to support them in performing daily life activities that most determine their ability to live independently. In fact, these activities cover two of the domains defined by IADLs [3]. To sustain people independence in execution of kitchen activities, it is necessary to provide technologies able to make the environment universally accessible and usable. This means that the environment should be able to support, as much as possible,

different users (i.e., normal people, disabled people, elderly, children, etc.) in order to ensure to all of them the possibility to achieve their goals, regardless of what are their physical or cognitive abilities. The achievement of this object is a very big challenge. As a matter of fact, technology is day by day the tool to support and ensure a better liveability including the kitchen activities, safety, comfort and well-being. In particular, it is necessary to develop a system that allows to coordinate the appliances and kitchen subsystems in order to optimize human-machine and human environment interaction both from physical and cognitive point of view. It should be able to support end-users, providing the necessary information according to the goal that user aims to achieve and support user's skill in performing such tasks. Moreover, it should be able to identify user behaviour, detect any possible modifications and modify itself accordingly.

In the last several years, the research on smart environments proposed new assistive systems that support and enhance the abilities of its occupants in executing domestic tasks. Most of these studies focused often only on services provided of smart home and marginally consider the end user preferences and applications. In particular, some solutions have been proposed that aim to support the user during the meal preparation [4,5], or during appliances interaction [6,7] or that aim to empower the user about environmental energy consumption [8]. Moreover, these studies do not address universally valid solutions: they neglect the adaptive solutions definition focusing on solutions for specific user profiles. To design product able to support different users with several disabilities, it is necessary to adopt an Ability-oriented Design perspective [9]. This approach encourages the design of self-adaptive or user-adaptable system interfaces, to provide the best possible match to user' abilities.

In this context, we propose a new adaptive smart kitchen environment able to support normal people and well as users with several typologies of impairment (i.e., visual, motor, cognitive) in performing cooking activities. This smart system, in particular, allows to manage the appliances and kitchen subsystems in order to optimize and improve usability for different contexts of use. The system consists of several devices (that is oven, dishwasher and the fridge) furnished with smart functions. The network is realized through wireless technology and it supplies communication from/to the gateway and from/to any appliance or device. The home automation platform is the 'core' of smart kitchen that organizes and elaborates all the data collected by the devices and operates events. It is a platform in a cloud service that gives network storage service and can be accessed from logic network applications. The system is

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managed through an adaptive web-user interface, which gives information on the features and usability of all kitchen available devices, allows to set and control all household appliances in a simple and intuitive way and gives information and alerts in case of the raise of uncommon situations requiring warning, and in any how adapting itself to the end-user's capabilities and needs.

2. THE ADAPTIVE INTERFACE

The interface system architecture is based on 4 main units, continuously communicating between each other: the database, the core, the interface and the doctor module. The interface system architecture is showed in Figure 3.

A Database Management System (DBMS) is designed to achieve a large set of structured data inputs and processes the amount of data requested by numerous users; this unit is in charge to collect the data arising from the different input of the system. In particular, the information is structured in four semantic areas. User Features Profile includes the User Personal Information and provides the description of the user's profile pattern, according to its cognitive and physical structure, status and preferences. The user's profile pattern is outline according to the coding provided by the International Classification of Functioning, Disability and Health [10]. This information is collected through an on-line analysis of the psycho-physical state of the user, according to a medical protocol defined by specialized research institutes. The User Features Profile it is populated by the information inserted in the doctor module: personal data and ICF. User Use Profile includes the User Model Context Information: previous interaction's history, user's preferences and information needs. Log Adaptation Actions represents the collection of all adaptation actions performed by the interface. It receives the information each time that the system performs an adaptation action: this information is necessary to control the system's adaptation degree and simultaneously the user's skills improvement in the process. Context Data, upon which are gathered all relevant data for the context user definition of the system. It shall record the information derived from environmental sensors (spatial context) associated with temporal coordinate. It will record the information from the sensors and smart objects in the environment. The core Module represents overall adaptive system pivot: it is composed of two adaptive mechanisms and a

changes monitoring system. Adaptable Engine shall make the system more suitable to the user profile. Adaptability is based on features and preferences that are known at a first interaction, and they are assumed to remain static during a single session of interaction. This engine takes as input the collected information in the User Profile Features to adapt the graphical interface features, such as text, size and type of font. The technical accessibility requirements and the corresponding control points for compliance verification have been defined on the basis of the Standards, Guidelines and success criteria contained in the Recommendation that the World Wide Web Consortium (W3C) - Web Accessibility Initiative (WAI) published on 11 December 2008 and which contains the Web Content Accessibility Guidelines 2.0 (WCAG 2.0).

Adaptive Engine shall make the system adaptive according to the use profile. Adaptivity is based on change mechanisms which include all dynamic features, such as interaction story based preferences, information contents, icons, layout, etc. The working systems take advantages of Bayesian Network adaptation mechanism, by using software called Netica. Bayesian networks [11] are one of the most complete and consistent knowledge tools for the acquisition, representation and use in uncertainty conditions, and the simulation software Netica, produced by Norsys Software Corporation [12], is a comprehensive tool for working with belief networks and influence diagrams. The Adaptive Engine depends on information by User Features, User Use and Context Data database, and apply changes on the Interface level: Normal Interface or Wizard Interface. Change monitoring system allows to generate two categories of alert: User Preferences Changes takes into account the user preferences changes, in terms of both graphics and content; User Abilities Changes takes into account the interface changes (both graphical and content), store them and send an alert when they frequently change, both in a positive and negative way, going to update the User Features Profile.

The monitoring system, placed at this level, is a tool to evaluate the psychophysical state and the disorder progress or regression over time. This assessment will be performed automatically by a machine. Finally, the interface module allows enabling system interaction and communication with user.

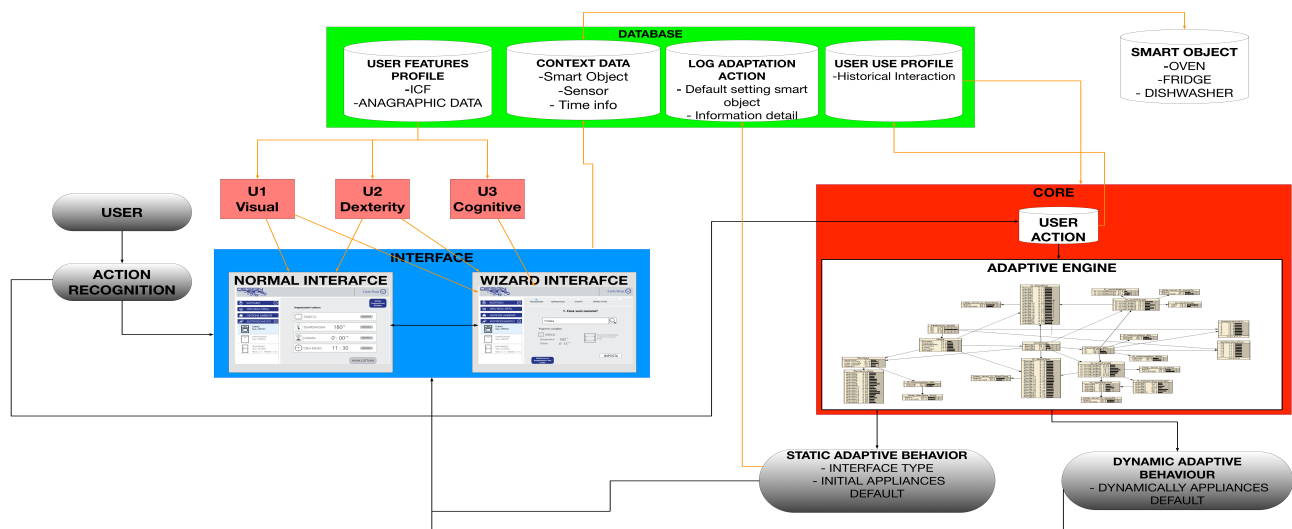


Figure 3. The Interface System Architecture

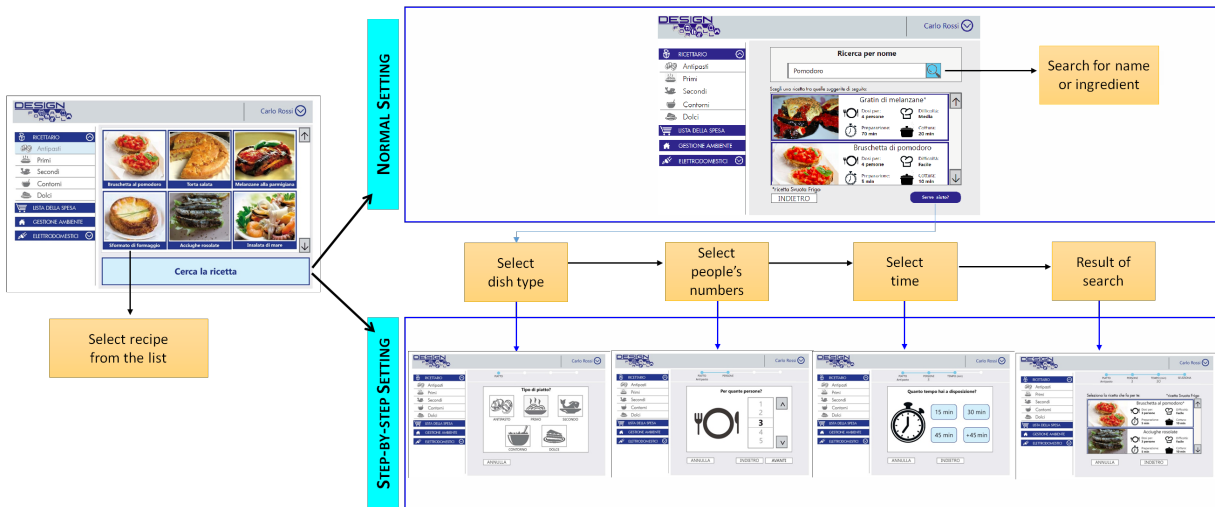


Figure 4. Select Recipe menu

Furthermore, the Adaptive Engine provides two different output: the static output sets the interface type (normal or wizard) and the initial appliances default, it is managed by probability calculating associated at the User Use Profile finding; the dynamic output refers to user action at the runtime. It manages the dynamically appliances default and update the interface at runtime. The interface is one of the most important modules of the entire architecture; this enables the system interaction and communication with the user. The Interface structure can be summarized in the following two aspects: graphic features, basic, i.e. standard features uniquely related to a disorder (colour blindness, visual disturbances, etc.), and advanced features that represent all dynamic features about adapted interface items according to specific residual function consequent to a specific disorder and they are designed on a single user. Contents represent all interface items editable according to user's actions and the user acts on the interface with his own preferences and needs. In particular, the needs related to specific deficit, visual actions, cognitive (procedural memory) and motor, according to user-defined profiles, it must be considered by the features interface. In this case, the technology must be of a "compensatory" and cooperate with the user in order to allow the activities of daily living independently. For instance, considering a kitchen, if the user decides to enter in the oven section, the system will offer in evidence more used recipes. To this end, the interface supports the following functional areas: 1) Meal preparation, 2) User interaction-appliance and 3) Environmental Comfort. With the aim to support the end user to set cooking program execution two different modes of information presentation were assumed: using a common menu (Normal Setting) and through setting driven (Step-by-step Setting or Wizard mode) process. The Wizard mode is designed to accomplish the task and minimizing the amount of information that the user should understand and manage. Consequently, this solution is suitable for users who have not familiarity with technology and / or have some cognitive dysfunction. On the other side, the Normal mode, is designed to support user without cognitive dysfunction and characterized by a good technology attitude.

Consequently, on first use, the setting menu will be presented in a "Wizard mode" to person with cognitive dysfunction, while in

"normal" mode for other profile. Figure 4 show an example of recipe setting menu. As one can observe, during interaction with the interface, the user can change the information presentation mode by tapping on the proper button. In detail, when the probability of default mode information exceeds a certain threshold, the information mode presented by default for that specific user is changed. Depending on user preferences, acquired in the profile phase acquisition, the system will provide a specific interactions mode. At any time, the user may change the interaction mode, changing preferences to his profile.

3. EXPERIMENTAL RESULTS

A preliminary usability evaluation has been carried out to assess system in terms of usability for each user target category. This first evaluation aimed to measure usability perceived by users during the first use of smart system: each user is asked to use the interface optimized, depending on his/her profile, according to default specification. Therefore, normal interface was presented to user with visual and dexterity problem, respectively adapted to maximize accessibility, while users with cognitive disability interacted with interface in wizard modality.

To recruit people, three users' profiles are considered, according to target user categories:

- Profile 1: aged between 45 and 54 years, with moderate visual impairment (retinopathy) and good attitude and skill with technology;
- Profile 2: adults aged between 55 and 65 years with moderate loss of dexterity (rheumatoid arthritis) and low attitude and skill with technology;
- Profile 3: adults over 70 years with mild to moderate cognitive impairment (dementia) which affect procedural memory and scarce skill with technology.

Five users per profile have been involved in the tests.

The usability lab was equipped with the interconnected smart appliances (i.e., oven, a dishwasher, a refrigerator), a table, a chair and a tablet. Ingredients were in the refrigerator or on the worktop, as well as the necessary tools, pots and dishes. Usability tests have been based on a task analysis. Users were asked to simulate the preparation of a baked omelette for two people, and in particular to perform four tasks: 1) find the recipe 2) retrieve the necessary ingredients and simulate the

preparation, 3) programming and start oven and 4) fill, set and start the dishwasher.

To support data collection both direct observation and Video Interaction Analysis (VIA) have been used. The following objective parameters have been collected for each task: % Completion without support (%C), Number of errors made by user during the interaction and (E) and Number of support requests (S). Before starting test session, the interface has been shown to the users and the various system functions were explained. At the end of the session, users were asked to answer to the SUS questionnaire [13].

Table 1. Usability test results

		Task 1	Task 2	Task 3	Task 4	SUS Score
Profile 1	C%	100%	100%	100%	100%	95,5
	E	0	0	1	0	
	S	0	0	0	0	
Profile 2	C%	100%	100%	60%	100%	83,5
	E	0	0	3	0	
	S	0	0	2	0	
Profile 3	C%	80%	100%	100%	100%	91
	E	1	0	1	1	
	S	1	0	0	0	

By analysing the results, we can observe that the propose smart interface can be considered very usable for all considered user profile. With respect to the Profile 1 and 3, there were no significant usability issues: the smart interface in normal modality has proven to be suitable for people with vision problems and to able to provide them support during preparation of meals activities, household appliances setting. At the same time, smart interface in wizard modality can effectively support people with problems in procedural memory. For this category, only a minor usability problem was observed during the execution of task 1: a user at the beginning found difficult perform tapping in a correct manner. However, it must point out that a total of 5 subjects (i.e., 2 belonging to Profile 2 and 3 to Profile 3) had never used a tablet before. Regard to Profile 2, we can observe that 2 users of 5 found difficult to set the oven with interface in normal modality. This can be due to the high number of operations that the user must perform to set the oven as well as the low users' technological attitude and skill. One possible solution is to provide the interface in wizard modality even for those users with poor skills in the use of technologies.

4. CONCLUSIONS & FUTURE WORKS

This paper presents the design of an innovative home automation system which aims to support users with visual, cognitive and motor related impairments in performing cooking activities through a smart adaptive interface. The results of a qualitative experimentation with final users has shown that the proposed interface is accessible and highly usable for users with some limitations such elderly with mild to moderate dementia and adult persona with moderate retinopathy and rheumatoid arthritis. In general, it emerged that the new smart kitchen system is able to improve the usability of household appliances, such as oven and dishwasher, as regards the programming and controlling operations. At the same time, it shows how the introduction of smart technologies in the kitchen environment can support users also with scarce attitude to technology.

The work is not definitive: the new smart kitchen has to be evaluated with a greater number of users in order to verify this first qualitative evaluation. Moreover, it will be necessary to

carry out a field study to assess the ability of adaptable features of interface to improve the system usability, as to assess the effectiveness of system to enhance the users' skills related to cooking activities and to improve users' independent living. An additional effort must be focused on the assessment of effectiveness of such system to support a doctor in monitoring patients' cognitive status.

5. ACKNOWLEDGMENTS

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